```
  this.seed ←
seed
#4
  this.numBits ← -ceil(maxS * ln(maxTolerance) / ln(2) /
ln(2))               #5
  if numBits > MAX_SIZE
then                                                  #6
    throw new
Error("Overflow")
  #7
  this.numHashFunctions ← -ceil(Math.log(maxTolerance) /
Math.log(2))          #8
  numElements ← ceil(numBits /
BITS_PER_INT)                                    #9
  this.bitsArray ← 0 (" i ∈ {0, …, numElements-1})
                        #10
  this.hashFunctions ← initHashFunctions(numHashFunctions,
maxSize)           #11
```

#1 Signature of the constructor method. Argument `maxTolerance` has a default value of 0.01, `seed` is by default initialized to a random integer. Not all programming languages provide an explicit syntax for default values in function signatures, but there are workarounds to cope with those that don't.

#2 Initially no element is stored in the filter, so size is initialized to `0`.

#3 We store in a class variable the (local) argument for the constructor. Same for next line.

#5 Compute the optimal number of bits needed: `m = - n * ln(p) / ln(2)`^{2}; `ceil(x)` is the standard ceiling function returning the smallest integer larger or equal to `x`.

#6 Check that the size will fit in memory without issues.